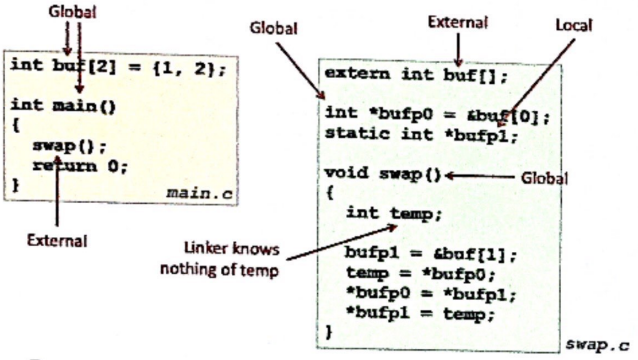
Three Kinds of Object Files (Modules)

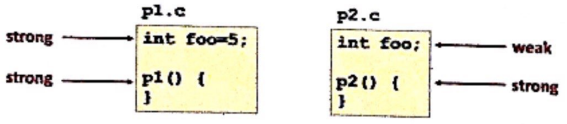
- **Relocatable object file (`.o` file)**
  - Contains code and data in a form that can be combined with other relocatable object files to form executable object file.
    - Each `.o` file is produced from exactly one source (`.c`) file
- **Executable object file (`a.out` file)**
  - Contains code and data in a form that can be copied directly into memory and then executed.
- **Shared object file (`.so` file)**
  - Special type of relocatable object file that can be loaded into memory and linked dynamically, at either load time or run-time.
  - Called *Dynamic Link Libraries* (DLLs) by Windows

Linker Symbols

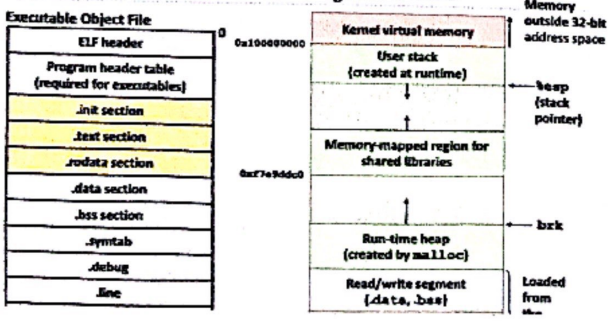
- **Global symbols**
  - Symbols defined by module *m* that can be referenced by other modules.
  - E.g.: non-`static` C functions and non-`static` global variables.
- **External symbols**
  - Global symbols that are referenced by module *m* but defined by some other module.
- **Local symbols**
  - Symbols that are defined and referenced exclusively by module *m*.
  - E.g.: C functions and variables defined with the `static` attribute.
  - **Local linker symbols are not local program variables**



- **Program symbols are either strong or weak**
  - ***Strong***: procedures and initialized globals
  - ***Weak***: uninitialized globals

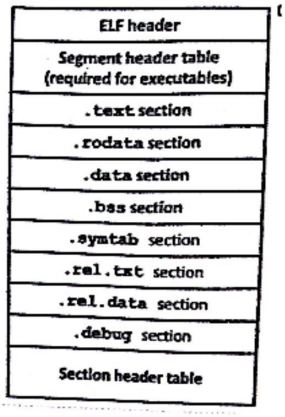


Loading Executable Object Files

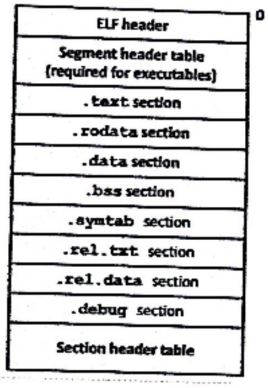


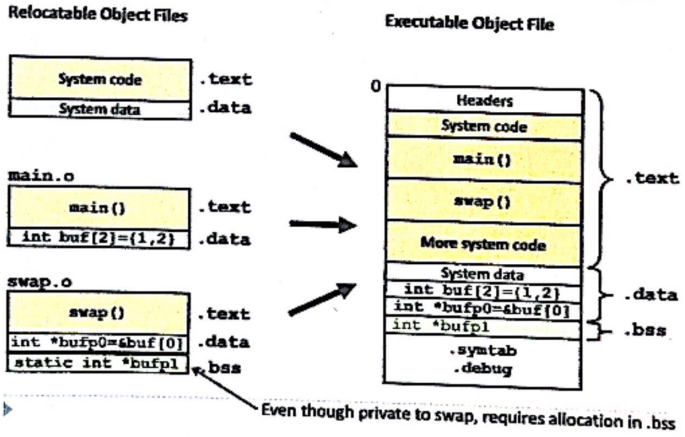
ELF Object File Format

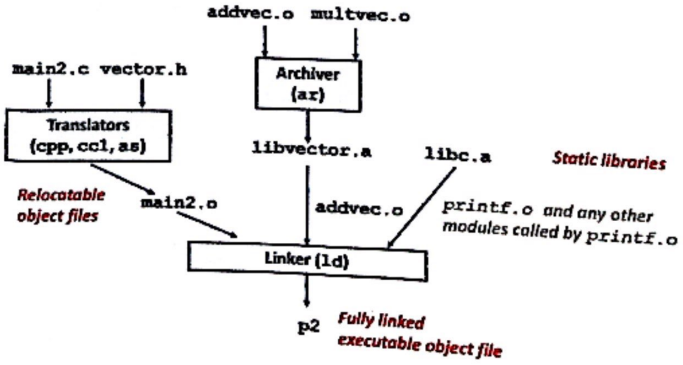
- **Elf header**
  - Word size, byte ordering, file type (.o, exec, .so), machine type, etc.
- **Segment header table**
  - Page size, virtual addresses memory segments (sections), segment sizes.
- **`.text` section**
  - Code
- **`.rodata` section**
  - Read only data: jump tables, ...
- **`.data` section**
  - Initialized global variables
- **`.bss` section**
  - Uninitialized global variables
  - "Block Started by Symbol"
  - "Better Save Space"
  - Has section header but occupies no space



- **`.symtab` section**
  - Symbol table
  - Procedure and static variable names
  - Section names and locations
- **`.rel.text` section**
  - Relocation info for `.text` section
  - Addresses of instructions that will need to be modified in the executable
  - Instructions for modifying.
- **`.rel.data` section**
  - Relocation info for `.data` section
  - Addresses of pointer data that will need to be modified in the merged executable
- **`.debug` section**
  - Info for symbolic debugging (`gcc -g`)
- **Section header table**
  - Offsets and sizes of each section







Some Arithmetic Operations

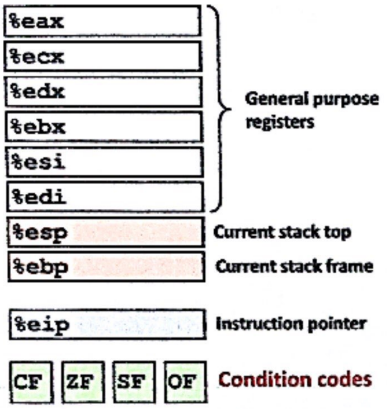
- Two Operand Instructions:

Format	*Computation*		
addl	*Src,Dest*	Dest = Dest + Src	
subl	*Src,Dest*	Dest = Dest – Src	
imull	*Src,Dest*	Dest = Dest * Src	
sall	*Src,Dest*	Dest = Dest << Src	*Also called shll*
sarl	*Src,Dest*	Dest = Dest >> Src	*Arithmetic*
shrl	*Src,Dest*	Dest = Dest >> Src	*Logical*
xorl	*Src,Dest*	Dest = Dest ^ Src	
andl	*Src,Dest*	Dest = Dest & Src	
orl	*Src,Dest*	Dest = Dest | Src	

- Watch out for argument order!
- No distinction between signed and unsigned int (why?)

Processor State (IA32, Partial)

- Information about currently executing program
  - Temporary data (`%eax`, …)
  - Location of runtime stack (`%ebp`,`%esp`)
  - Location of current code control point (`%eip`, …)
  - Status of recent tests (CF, ZF, SF, OF)



Jumping

- jX Instructions
  - Jump to different part of code depending on condition code

jX	Condition	Description
jmp	1	Unconditional
je	ZF	Equal / Zero
jne	~ZF	Not Equal / Not Zero
js	SF	Negative
jns	~SF	Nonnegative
jg	~(SF^OF)&~ZF	Greater (Signed)
jge	~(SF^OF)	Greater or Equal (Signed)
jl	(SF^OF)	Less (Signed)
jle	(SF^OF)|ZF	Less or Equal (Signed)
ja	~CF&~ZF	Above (unsigned)
jb	CF	Below (unsigned)

Bad Cases for Conditional Move

Expensive Computations

```
val = Test(x) ? Hard1(x) : Hard2(x);
```

- Both values get computed
- Only makes sense when computations are very simple

Risky Computations

```
val = p ? *p : 0;
```

- Both values get computed
- May have undesirable effects

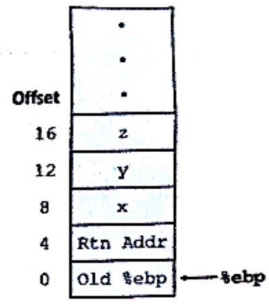
Computations with side effects

```
val = x > 0 ? x*=7 : x+=3;
```

- Both values get computed

Understanding `arith`

```
int arith(int x, int y, int z)
{
  int t1 = x+y;
  int t2 = z+t1;
  int t3 = x+4;
  int t4 = y * 48;
  int t5 = t3 + t4;
  int rval = t2 * t5;
  return rval;
}
```




```
movl   8(%ebp), %ecx
movl   12(%ebp), %edx
leal   (%edx,%edx,2), %eax
sall   $4, %eax
leal   4(%ecx,%eax), %eax
addl   %ecx, %edx
addl   16(%ebp), %edx
imull  %edx, %eax
```

Condition Codes (Implicit Setting)

- Single bit registers
  - CF Carry Flag (for unsigned) SF Sign Flag (for signed)
  - ZF Zero Flag OF Overflow Flag (for signed)
- Implicitly set (think of it as side effect) by arithmetic operations

  Example: `addl`/`addq` *Src,Dest* ↔ `t = a+b`

  CF set if carry out from most significant bit (unsigned overflow)

  ZF set if `t == 0`

  SF set if `t < 0` (as signed)

  OF set if two's-complement (signed) overflow

  `(a>0 && b>0 && t<0) || (a<0 && b<0 && t>=0)`
- Not set by `lea` instruction

Conditional Branch Example (Cont.)

```
int goto_ad(int x, int y)
{
  int result;
  if (x <= y) goto Else;
  result = x-y;
  goto Exit;
Else:
  result = y-x;
Exit:
  return result;
}
```

```
absdiff:
   pushl  %ebp              } Setup
   movl   %esp, %ebp
   movl   8(%ebp), %edx     } Body1
   movl   12(%ebp), %eax
   cmpl   %eax, %edx
   jle    .L6
   subl   %eax, %edx        } Body2a
   movl   %edx, %eax
   jmp .L7
.L6:
   subl %edx, %eax          } Body2b
.L7:
   popl %ebp                } Finish
   ret
```

Shared Libraries

- Static libraries have the following disadvantages:
  - Duplication in the stored executables (every function need std libc)
  - Duplication in the running executables
  - Minor bug fixes of system libraries require each application to explicitly relink
- Modern solution: Shared Libraries
  - Object files that contain code and data that are loaded and linked into an application *dynamically*, at either *load-time* or *run-time*
  - Also called: dynamic link libraries, DLLs, `.so` files

Jump Table Structure

Switch Form

```
switch(x) {
  case val_0:
    Block 0
  case val_1:
    Block 1
    • • •
  case val_n-1:
    Block n-1
}
```

Jump Table

```
jtab: Targ0
      Targ1
      Targ2
      •
      •
      •
      Targn-1
```

Jump Targets

```
Targ0:   Code Block 0
Targ1:   Code Block 1
Targ2:   Code Block 2
   •
   •
   •
Targn-1: Code Block n-1
```

Approximate Translation

```
target = JTab[x];
goto *target;
```

Shared Libraries (cont.)

- Dynamic linking can occur when executable is first loaded and run (load-time linking).
  - Common case for Linux, handled automatically by the dynamic linker (`ld-linux.so`).
  - Standard C library (`libc.so`) usually dynamically linked.
- Dynamic linking can also occur after program has begun (run-time linking).
  - In Linux, this is done by calls to the `dlopen()` interface.
    - Distributing software.
    - High-performance web servers.
- Shared library routines can be shared by multiple processes.
  - More on this when we learn about virtual memory

Switch Statement Example (IA32)

```
long switch_eg(long x, long y, long z)
{
    long w = 1;
    switch(x) {
      . . .
    }
    return w;
}
```

Jump table

```
.section    .rodata
  .align 4
.L7:
  .long     .L2 # x = 0
  .long     .L3 # x = 1
  .long     .L4 # x = 2
  .long     .L5 # x = 3
  .long     .L2 # x = 4
  .long     .L6 # x = 5
  .long     .L6 # x = 6
```

Setup:

```
switch_eg:
   pushl   %ebp            # Setup
   movl    %esp, %ebp      # Setup
   movl    8(%ebp), %eax   # eax = x
   cmpl    $6, %eax        # Compare x:6
   ja      .L2             # If unsigned > goto default
Indirect jump  jmp  *.L7(,%eax,4) # Goto *JTab[x]
```

Assembly Setup Explanation

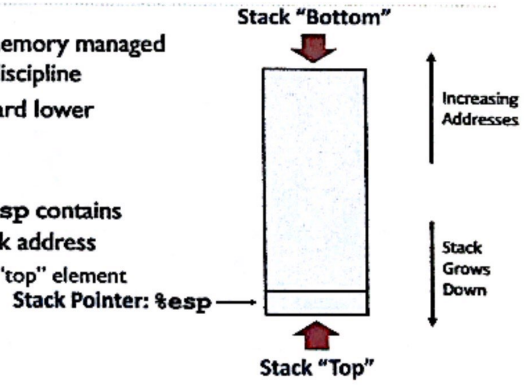
- Table Structure
  - Each target requires 4 bytes
  - Base address at `.L7`
- Jumping
  - Direct: `jmp .L2`
  - Jump target is denoted by label `.L2`
  - Indirect: `jmp *.L7(,%eax,4)`
  - Start of jump table: `.L7`
  - Must scale by factor of 4 (labels have 32-bits = 4 Bytes on IA32)
  - Fetch target from effective Address `.L7 + eax*4`
    - Only for $0 \le x \le 6$

Jump table

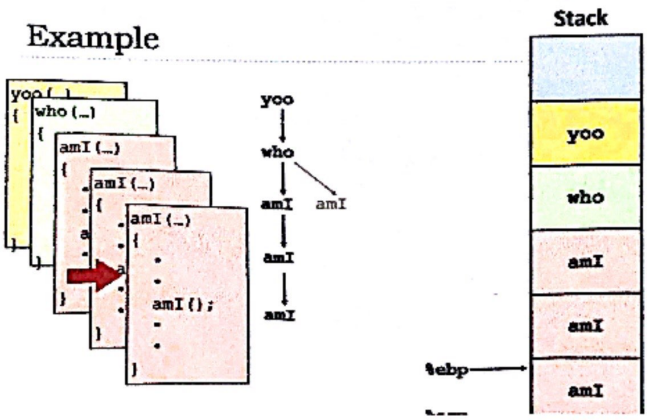
```
.section    .rodata
  .align 4
.L7:
  .long     .L2 # x = 0
  .long     .L3 # x = 1
  .long     .L4 # x = 2
  .long     .L5 # x = 3
  .long     .L2 # x = 4
  .long     .L6 # x = 5
  .long     .L6 # x = 6
```

IA32 Stack

- Region of memory managed with stack discipline
- Grows toward lower addresses
- Register `%esp` contains lowest stack address
  - address of "top" element



Example



IA32/Linux+Windows Register Usage

- `%eax, %edx, %ecx`
  - Caller saves prior to call if values are used later
- `%eax`
  - also used to return integer value
- `%ebx, %esi, %edi`
  - Callee saves if wants to use them
- `%esp, %ebp`
  - special form of callee save
  - Restored to original values upon exit from procedure